

● MEDIUM

● ALGEBRA

● ANSWER KEY

Algebra

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Q1**49**

The correct answer is 49. It's given that the equation $7g + 7b = 840$ represents the number of blue tiles, b , and the number of green tiles, g , an artist needs for an 840-square-inch tile project. It's also given that the artist needs 71 blue tiles for the project. Substituting 71 for b in the equation $7g + 7b = 840$ yields $7g + 7(71) = 840$, or $7g + 497 = 840$. Subtracting 497 from both sides of this equation yields $7g = 343$. Dividing both sides of this equation by 7 yields $g = 49$. Therefore, the artist needs 49 green tiles for the project.

Q2**34**

The correct answer is 34. It's given that a line segment has a length of 115 cm and is divided into three parts, where one part is 47 cm long and the other two parts have lengths that are equal. If x represents the length, in cm, of each of the two parts of equal length, then the equation $47 + x + x = 115$, or $47 + 2x = 115$, represents this situation. Subtracting 47 from each side of this equation yields $2x = 68$. Dividing each side of this equation by 2 yields $x = 34$. Therefore, the length, in cm, of one of the two parts of equal length is 34.

Q3

C

C. $y = 8x + 2$

Choice C is correct. It's given that the relationship between x and y is linear. An equation representing a linear relationship can be written in the form $y = mx + b$, where m is the slope and b is the y -coordinate of the y -intercept of the graph of the relationship in the xy -plane. It's given that for every increase in the value of x by 1, the value of y increases by 8. The slope of a line can be expressed as the change in y over the change in x . Thus, the slope, m , of the line representing this relationship can be expressed as $\frac{8}{1}$, or 8. Substituting 8 for m in the equation $y = mx + b$ yields $y = 8x + b$. It's also given that when the value of x is 2, the value of y is 18. Substituting 2 for x and 18 for y in the equation $y = 8x + b$ yields $18 = 8(2) + b$, or $18 = 16 + b$. Subtracting 16 from each side of this equation yields $2 = b$. Substituting 2 for b in the equation $y = 8x + b$ yields $y = 8x + 2$. Therefore, the equation $y = 8x + 2$ represents this relationship.

Choice A is incorrect. This equation represents a relationship where for every increase in the value of x by 1, the value of y increases by 2, not 8, and when the value of x is 2, the value of y is 22, not 18.

Choice B is incorrect. This equation represents a relationship where for every increase in the value of x by 1, the value of y increases by 2, not 8, and when the value of x is 2, the value of y is 12, not 18.

Choice D is incorrect. This equation represents a relationship where for every increase in the value of x by 1, the value of y increases by 3, not 8, and when the value of x is 2, the value of y is 32, not 18.

Q4

B

B. 5

Choice B is correct. Adding the first equation to the second equation in the given system yields $x + 2y + x - 2y = 6 + 4$, or $x + x + 2y - 2y = 10$. Combining like terms in this equation yields $2x = 10$. Dividing both sides of this equation by 2 yields $x = 5$. Thus, the value of x is 5.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $2x$, not x .

Q5

D

D.

x	y
3	21
5	35
8	56

Choice D is correct. A solution (x, y) to the given inequality is a value of x and the corresponding value of y such that the value of y is greater than the value of $7x - 4$. All the tables in the choices have the same three values of x , so each of the three values of x can be substituted in the given inequality to compare the corresponding values of y in each of the tables. Substituting 3 for x in the given inequality yields $y > 7(3) - 4$, or $y > 17$. Substituting 5 for x in the given inequality yields $y > 7(5) - 4$, or $y > 31$. Substituting 8 for x in the given inequality yields $y > 7(8) - 4$, or $y > 52$. Therefore, when $x = 3$, $x = 5$, and $x = 8$, the corresponding values of y must be greater than 17, greater than 31, and greater than 52, respectively. In the table in choice D, when $x = 3$, the corresponding value of y is 21, which is greater than 17; when $x = 5$, the corresponding value of y is 35, which is greater than 31; when $x = 8$, the corresponding value of y is 56, which is greater than 52. Of the given choices, only choice D gives values of x and their corresponding values of y that are all solutions to the given inequality.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q6**C**

C. The total number of cups of jam in the large containers

Choice C is correct. It's given that the equation $4x + 6y = 36$ represents the situation where Lily filled x small containers and y large containers with all the jam she made, which was 36 cups. Therefore, $6y$ represents the total number of cups of jam in the large containers.

Choice A is incorrect. The number of large containers Lily filled is represented by y , not $6y$.

Choice B is incorrect. The number of small containers Lily filled is represented by x , not $6y$.

Choice D is incorrect. The total number of cups of jam in the small containers is represented by $4x$, not $6y$.

Q7**C**

C. Infinitely many

Choice C is correct. If the two sides of a linear equation are equivalent, then the equation is true for any value. If an equation is true for any value, it has infinitely many solutions. Since the two sides of the given linear equation $66x = 66x$ are equivalent, the given equation has infinitely many solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8**67**

The correct answer is 67. It's given that $f5 = 32$. Therefore, for the given function f , when $x = 5$, $fx = 32$. Substituting 5 for x and 32 for fx in the given function $fx = 4x + kx - 1$ yields $32 = 4(5) + k(5) - 1$, or $32 = 20 + 5k$. Subtracting 20 from each side of this equation yields $12 = 5k$. Dividing each side of this equation by 5 yields $k = \frac{12}{5}$. Substituting $\frac{12}{5}$ for k in the given function $fx = 4x + kx - 1$ yields $fx = 4x + \frac{12}{5}x - 1$, which is equivalent to $fx = 4x + \frac{12}{5}x - \frac{5}{5}$, or $fx = \frac{20x + 12x - 5}{5}$. Substituting 10 for x into this equation yields $f10 = \frac{20(10) + 12(10) - 5}{5}$, or $f10 = \frac{200 + 120 - 5}{5}$. Therefore, the value of $f10$ is 67.

Q9**C**

C. $-12x + 20y = 52$

Choice C is correct. It's given that the system has infinitely many solutions. A system of two linear equations has infinitely many solutions when the two linear equations are equivalent. Dividing both sides of the given equation by 5 yields $-3x + 5y = 13$. Dividing both sides of choice C by 4 also yields $-3x + 5y = 13$, so choice C is equivalent to the given equation. Thus, choice C could be the second equation in the system.

Choice A is incorrect. The system consisting of this equation and the given equation has one solution, not infinitely many solutions.

Choice B is incorrect. The system consisting of this equation and the given equation has one solution, not infinitely many solutions.

Choice D is incorrect. The system consisting of this equation and the given equation has no solution, not infinitely many solutions.

Q10

A

A. $120 \leq H \leq 162$

Choice A is correct. When Alice trains, it's recommended that p be between 0.5 and 0.85. Therefore, her target training heart rate is represented by the values of H corresponding to $0.5 \leq p \leq 0.85$. When $p = 0.5$, $H = 120(0.5) + 60$, or $H = 120$. When $p = 0.85$, $H = 120(0.85) + 60$, or $H = 162$. Therefore, the inequality that describes Alice's target training heart rate is $120 \leq H \leq 162$.

Choice B is incorrect. This inequality describes Alice's target heart rate for $0.35 \leq p \leq 0.5$.

Choice C is incorrect. This inequality describes her target heart rate for $0 \leq p \leq 0.85$. Choice D is incorrect. This inequality describes her target heart rate for $0 \leq p \leq 0.35$.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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